Project Report

# Machine Learning-Based Insurance Risk Assessment Pipeline for Natural Disaster Prediction and Premium Estimation


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# 1. Introduction

## 1.1 What the application does

This project is a small web application that connects two ideas. The first idea is how likely a US state is to receive a major disaster declaration from FEMA in a given year. The second idea is how much a person's health insurance charges are expected to be. The application predicts both and shows them side by side, so a user can see a state's disaster risk next to a predicted insurance cost for a sample person.

The project is a research proof of concept. The interface itself says that it is not a real premium calculator. Its practical purpose is to show how a disaster risk score can be built from past records, how that score can be passed to an insurance cost model, and whether the link actually helps the cost model. The likely users are students, researchers and analysts who want to test this idea in a clear and honest way. It is not meant for pricing real policies.

## 1.2 The problem and the terms used

Insurers need to understand how natural disasters affect their business. Disasters can cause large losses, and the risk is not the same in every place. A simple way to include disaster risk in insurance work is to give each location a risk score and use it as an extra input when estimating costs. This project tries that approach with the data that was available to us.

Several money-related words are easy to mix up, so we use them in a fixed way in this report.

**Insurance charges.** This is the value in the column called charges in the insurance dataset [16]. It is the number our regression model predicts. The dataset and our code do not say whether it is a yearly premium or a total of claims, so we call it insurance charges and do not call it a premium or a claim.

**Premium.** A premium is the price a customer pays for cover. Our project has no premium formula. The two estimates shown in the interface are predicted charges, not premiums.

**Claim.** A claim is a request for payment after a loss. Our project has no claim data and does not predict claims.

**Disaster loss.** This means the money damage caused by a disaster. Our project does not use loss data and does not predict losses.

**Disaster declaration.** This is a major disaster declaration made by FEMA for a state. Our disaster model predicts whether a state will have at least one such declaration in the next year.

## 1.3 Scope and objectives

The title of the project mentions natural disaster prediction and premium estimation. The implemented system covers this in a narrower way. It does not forecast individual storms, floods or fires. It estimates, from the history of past declarations, the chance that a state is declared a disaster area in the next year. It also does not estimate premiums. It predicts

insurance charges from age, sex, BMI, number of children, smoking status and region. The two parts meet only through the region, as explained in Section 1.5.

The main objectives were as follows. First, turn the FEMA declaration history into a yearly risk estimate for each state. Second, predict insurance charges from personal details. Third, test whether adding the regional disaster risk improves the charges model. Fourth, present the results in a web interface that is open about its limits.

### 1.4 Main implemented features

The application has three tabs. The Estimator tab takes a state and a personal profile and returns the state's disaster risk, the region's disaster risk, a baseline charges estimate (Model A) and a disaster-aware charges estimate (Model B), together with their difference and a short explanation. The tab also has a what-if slider that shows how a third model (Model C) responds when the risk value changes. The Risk Map tab shows the predicted chance of a declaration for every state and the average for each region. The Model Lab tab shows the evaluation tables for both models. The interface checks the inputs and warns the user when the age is above the range seen in training.

### 1.5 How disaster risk and insurance estimation are connected

The two datasets do not share any rows. The insurance data has no state, date or disaster information. It has only a region with four possible values. To connect the parts, we assigned every state to one of the four regions using a rule written in the configuration file. This is our own assumption and not a feature of the data. The disaster model gives each state a probability. The probability values of the states in a region are averaged, and this average becomes a column called disaster_risk in the insurance data.

This design has a clear consequence. Every person in the same region receives the same disaster_risk value, so the new column carries no more location detail than the region name. The code measures this and reports it. We therefore treat the insurance part as a test of the idea, and we do not claim that disaster risk explains insurance charges. The same limit appears again in the methodology and results sections.

## 2. Related Work

We reviewed the fifteen papers listed for this project. They cover three groups of work. Some papers build machine learning models for disaster damage or insurance claims [1], [2], [6]-[8], [11], [12]. Some are reviews or actuarial models of catastrophe insurance [4], [5], [10], [13]. The rest are economic studies of how home insurance is priced when disaster risk is present [3], [9], [14], [15]. Each paper was checked against its publisher page, a repository copy or an indexing service. Table 1 lists the method, the main findings and the gaps that matter for our project.

Papers [3] and [14] are NBER working papers and have not gone through journal peer review. Paper [15] is a chapter in a conference volume published as a book series. The other

twelve are journal articles. Papers [1] and [11] were read in detail. For the others we relied on the abstract and the publisher record, because the full text was not available to us. In the table, limitations marked Stated come from the authors. Limitations marked Inferred are our own reading of how the paper relates to our project.

**Table 1: Literature review of the fifteen papers**

| Ref. | Paper, authors, year, type | Method or approach | Main contribution or findings | Gap or limitation for our project |
|---|---|---|---|---|
| [1] | Machine learning-based insurance risk assessment pipeline for natural disaster prediction and claims estimation.<br>A. H. Mujtaba, 2026.<br>*Journal article (Front. Artif. Intell.)* | Cascaded pipeline on 68,485 FEMA declaration records (1953-2025) with 49 engineered features. Classifies disaster type, estimates severity and claims with ensemble models, then gives a risk score and a premium multiplier from 0.8x to 1.7x. Loss and claim targets are synthetic. | Disaster type accuracy was 92.24% with a random split but 53.86% with a temporal hold-out. Claims R-squared was 0.97 on a random split but negative on a temporal hold-out. The author links this to the synthetic targets. | Stated: the synthetic targets include a 3% yearly inflation factor, which breaks temporal testing, and real loss data is needed.<br>Inferred: it predicts disaster type, not whether a state is declared next year, and its claims are synthetic, so it cannot be compared with our health charges data. |
| [2] | Predicting flood insurance claims with hydrologic and socioeconomic demographics via machine learning: Exploring the roles of topography, minority populations, and political dissimilarity.<br>J. Knighton et al., 2020.<br>*Journal article (J. Environ. Manage.)* | Random Forest regression and classification to predict parcel-level and tract-level flood insurance claims in New York State, using hydrologic and socioeconomic data. | Topography best described claim frequency. Adding socioeconomic data improved prediction at both scales. The classification models likely over-estimated the flood-exposed area. | Stated: classification likely over-estimated flood-exposed areas.<br>Inferred: one state and one hazard, so it says little about multi-hazard or national risk. |
| [3] | Property insurance and disaster risk: New evidence from mortgage escrow data.<br>B. J. Keys and P. Mulder, 2024 (revised).<br>*NBER working paper 32579* | Builds a new data set of home insurance premiums (over 74 million in the current version, 2014-2024) by splitting mortgage escrow payments, then studies how premiums relate to disaster risk. | Premiums rose quickly and the pass-through from disaster risk into premiums doubled. The authors tie the faster rise to a repricing of catastrophe risk in global capital markets. | Inferred: an economic study of existing premiums, not a prediction model. It shows that real premiums depend on fine location risk, while our data has only four regions. |
| [4] | The state of the art and future of climate risk insurance modeling.<br>M. W. Ingels et al., 2024.<br>*Journal article, review (Ann. N. Y. Acad. Sci.)* | Literature review that sorts models into catastrophe modelling, actuarial modelling and theoretical modelling. | More than half of the reviewed papers cover flood. Most forward-looking models find that climate change and social change raise future risk and premiums. | Stated: flood is over-represented, multi-hazard work is rare, fewer than half of the studies look forward with climate scenarios, and non-agricultural commercial insurance is under-studied.<br>Inferred: our project is multi-peril but uses only past declarations and no climate scenarios. |
| [5] | A discrete model of the expected loss for catastrophe insurance in natural disasters.<br>C. Huang and X. Shi, 2011.<br>*Journal article (J. Risk Anal. Crisis Response)* | Discrete actuarial model. It uses the probability of catastrophe events, a damage-ratio matrix and an insurance-loss matrix to compute expected loss and a premium rate. Applied to typhoon insurance for houses in Fujian Province, China. | The authors say the usual actuarial model is too theoretical and that their model is versatile, easy to use and gives more reliable results. | Inferred: a regional case study from 2011 that is not data-driven machine learning, and the abstract does not state its limits. |

| Ref. | Paper, authors, year, type | Method or approach | Main contribution or findings | Gap or limitation for our project |
|---|---|---|---|---|
| [6] | Learning inter-annual flood loss risk models from historical flood insurance claims.<br>J. Salas, A. Saha and S. Ravela, 2023.<br>*Journal article (J. Environ. Manage.). An earlier arXiv preprint has a longer title.* | Regression models on National Flood Insurance Program (NFIP) claims. The preprint compares a neural network, XGBoost and a Gaussian Process, adds rainfall data and corrects bias with a Burr distribution. | In the preprint, XGBoost gave the best results, bias correction and rainfall helped, and one regional study reached R-squared 0.807. The journal version reports that shifting or expanding the training window made little difference to short-term predictions. | Inferred: flood only. The numbers above are from the preprint and may differ from the journal version. |
| [7] | Predicting flood property insurance claims over CONUS, fusing big Earth observation data.<br>Q. Yang et al., 2022.<br>*Journal article (Bull. Am. Meteorol. Soc.)* | Random Forest model called iClaim. It joins NFIP records with flood extent, rainfall and river data. A two-step scheme first classifies damage level and then regresses the number of claims. | R-squared was over 0.5 for grid cells per event, over 0.9 for counties per event and over 0.95 for event totals. | Inferred: it predicts the number of claims, not their cost, and covers flood only. |
| [8] | AI meets the eye of the storm: Machine learning-driven insights for hurricane damage risk assessment in Florida.<br>S. Maha Arachchige and B. Pradhan, 2025.<br>*Journal article (Earth Syst. Environ.)* | Stacked ensemble machine learning model trained on building-level insurance data from 1985 to 2024 and remote sensing variables, to assess hurricane flood damage in Florida. | The model predicted flood damage at ZIP Code Tabulation Area level with a mean absolute error of 11.3%. Building value and the 1970s building code were important in the analysis. | Stated: insurance data may be an imperfect picture of Florida because of under-insurance and people who cannot afford insurance.<br>Inferred: single state, single peril. |
| [9] | Pricing of climate risk insurance: Regulation and cross-subsidies.<br>S. S. Oh, I. Sen and A.-M. Tenekedjieva, 2026.<br>*Journal article (J. Finance)* | Economic analysis using state rate-regulation rules in an instrumental variable design and a border discontinuity design on home insurance rates. | Insurers do not adjust rates in highly regulated states and raise rates in less regulated states instead. Rates and risks drift apart and insurers leave highly regulated states. | Inferred: it explains why real premiums can differ from risk. It is not a prediction method, and it shows that a pure risk-based premium is not what the market charges. |
| [10] | A critical review of hurricane risk assessment models and predictive frameworks.<br>S. Maha Arachchige, B. Pradhan and H.-J. Park, 2025.<br>*Journal article, review (Geosci. Front.)* | Systematic review of 94 articles that compares multi-criteria decision making, numerical models and machine learning for hurricane risk. | The top six predictors were land use, slope, precipitation, elevation, population density and soil texture or drainage. Random Forest was used in 42% of machine learning studies and neural networks in 19%. | Stated: most assessments rely on hurricane frequency rather than damage, indirect impacts are rarely considered, mitigation is used as a predictor in only 5% of studies, and model validation, especially transfer to a new region, is weak.<br>Inferred: our disaster model also works with declaration counts rather than damage. |
| [11] | A hybrid machine learning framework for predicting hurricane losses in parametric insurance with highly imbalanced data.<br>Y. Men et al., 2026.<br>*Journal article (Algorithms)* | Two-track framework. Classifiers sort events into non-payable, partly payable and fully payable. A regression model estimates the payout for partly payable events. XGBoost, LightGBM and CatBoost are used on simulated hurricane data for one hotel in Kingston, Jamaica. | Only 0.09% of the 28,698 test events reached a payable loss. The hybrid framework gave a better balance between detecting payable events and predicting the payout than the single regression models. | Stated: the framework is complex and hard to interpret and is a research benchmark, not for direct use in contracts.<br>Inferred: one asset and simulated data, not real claims. |

| Ref. | Paper, authors, year, type | Method or approach | Main contribution or findings | Gap or limitation for our project |
|---|---|---|---|---|
| [12] | Hurricane wind loss modeling using insurance claims data.<br>N. O. Tackie-Otoo et al., 2026.<br>*Journal article (Natural Hazards)* | Two models on policy and claims data from four hurricanes in eastern North Carolina. One XGBoost model predicts whether a building has a loss, and an XGBoost-Tweedie model predicts the loss amount. | For a future hurricane the models predicted the number of claims within 2% to 12% and the total loss within about 1% to 16%. Wind speed and total rain were the most important features. | Stated: the claims data are proprietary and cannot be shared.<br>Inferred: one region, and the results cannot be reproduced without the data. |
| [13] | Loss assessment and earthquake parametric insurance pricing using hybrid Bayesian network.<br>D. Bai and X. Chen, 2026.<br>*Journal article (Risk Analysis)* | Hybrid Bayesian network for fast and post-disaster earthquake loss assessment, plus a two-parameter pricing model that uses magnitude and epicentre intensity. | The network beat traditional parametric methods, multiple linear regression and Random Forest. Adding casualty data improved accuracy, and the pricing model can reduce basis risk. | Inferred: earthquake only, and parametric pricing is a different product from the cost estimate in our project. |
| [14] | Pricing protection: Credit scores, disaster risk, and home insurance affordability.<br>J. Blonz et al., 2026.<br>*NBER working paper 34848* | Links about 70 million home insurance policies to mortgages and property-level disaster risk, and uses a temporary ban on credit-based pricing in Washington State. | Credit scores affect premiums about as much as disaster risk. Homeowners with low credit pay 24% more for the same cover. | Inferred: shows that real premiums depend on factors other than disaster risk, which our charges model does not include. |
| [15] | The efficiency and equity impacts of risk classification in catastrophe insurance.<br>J. Boomhower and M. Fowlie, 2026.<br>*Book-series chapter (Environ. Energy Policy Econ., vol. 7)* | Empirical study of California home insurance with a focus on wildfire risk, comparing finer and coarser risk classification. | More detailed pricing raises costs a lot for the highest-risk properties and increases the use of cost-effective protection. Because wildfire risk is linked to lower income, it also moves more cost to lower-income households. | Inferred: it shows why fine location detail matters for risk-based pricing. Our four-region approach is far coarser. |

### 2.1 Common findings

Three themes appear across the papers. First, machine learning models, especially tree-based ensembles such as Random Forest and XGBoost, are the most common tools for predicting disaster damage or insurance claims [1], [2], [6], [7], [10]-[12]. Second, adding more information about the place, such as terrain, rainfall or social data, usually helps the models [2], [6], [7], [12]. Third, the pricing studies show that real insurance prices follow disaster risk only in part, and that regulation, capital markets and credit scores also matter [3], [9], [14], [15].

### 2.2 Remaining limitations

Many studies cover a single hazard or a single region [2], [6]-[8], [12], and the review papers note that multi-hazard work is rare [4], [10]. Several studies rely on claims data that are private or simulated, which makes results hard to reproduce [1], [11], [12]. Paper [1] also shows how random splits can give results that look much better than time-based tests. The reviews call for stronger validation, especially when a model is moved to a new region [10]. Finally, the pricing studies appear to rely on very large policy and mortgage data sets that a student project cannot easily obtain.

### 2.3 How our project relates to the literature

Our project is closest in scope to paper [1], whose title is similar to ours. Both use FEMA declaration data and both link disaster risk to an insurance quantity. The implementations are different. Paper [1] classifies the type of disaster, builds synthetic loss and claim targets and uses a rule-based premium multiplier. Our project predicts whether a state is declared a disaster area in the next year, and it predicts the charges in a public health insurance data set. We used time-based validation for the disaster model, which follows the warning in [1] about random splits. We did not use claims, losses or climate inputs, so we do not address the data gaps raised in [4], [10] and [12].

The pricing papers [3], [9], [14], [15] help to explain our main result. They show that useful risk pricing needs fine location data, such as ZIP code or property level. Our insurance data has only four regions, which is why the disaster feature could not add information. We do not claim that our project closes any gap in the literature. It is a small, transparent test of an idea that the literature suggests needs better data.

## 3. Methodology

### 3.1 Overview of the workflow

The system has two modelling paths that are trained in one script (train.py) and joined at one point. Figure 1 shows the whole process. Path 1 turns the FEMA declaration history into a yearly risk estimate for each state. Path 2 predicts insurance charges from personal details. The two paths connect where the average risk of each region is added to the insurance data as

the column disaster_risk. The trained models and tables are saved to disk, and a Flask web application loads them to answer user requests.

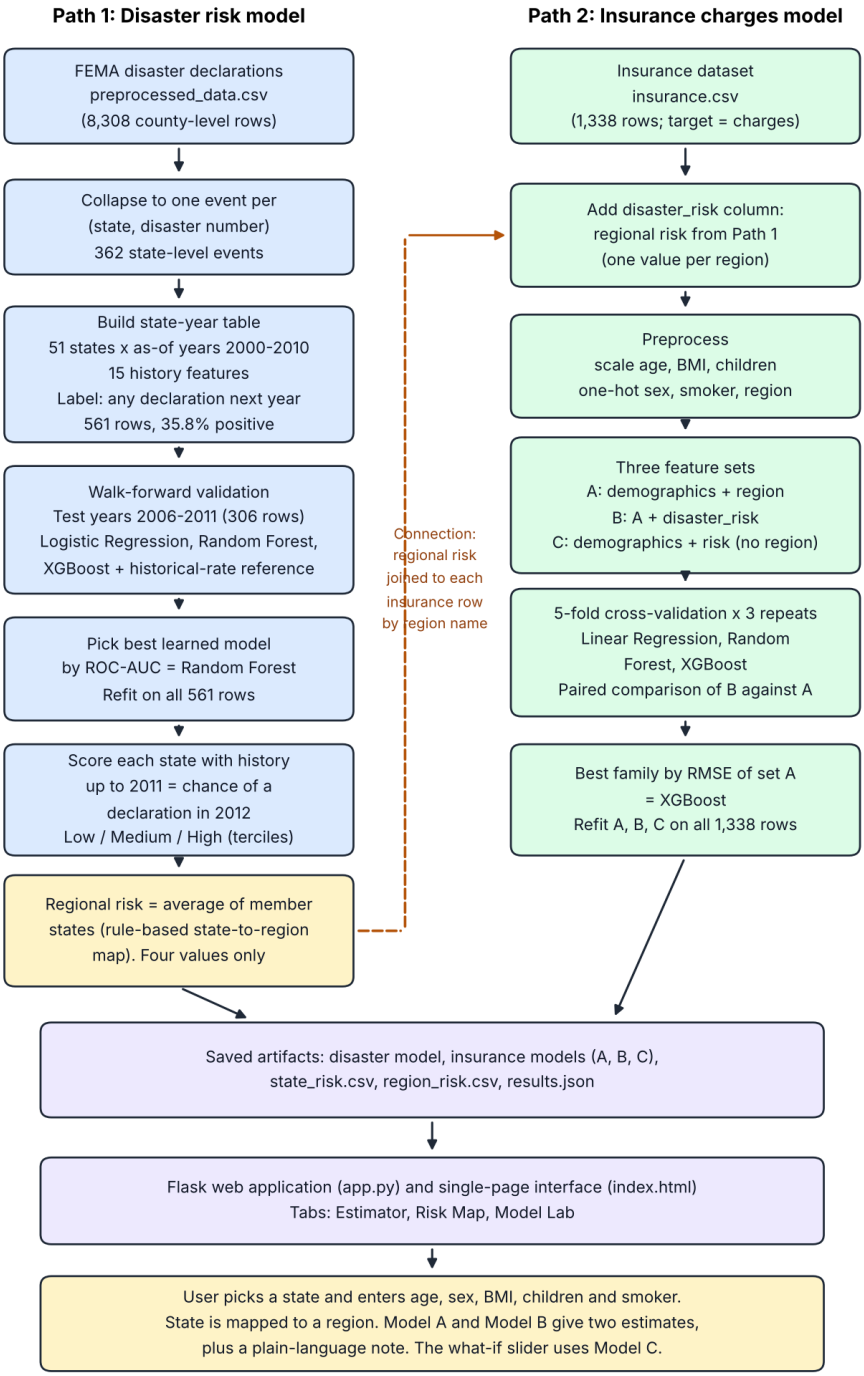



*Figure 1: Flow diagram of the implemented pipeline. Path 1 (blue) is the disaster risk model and Path 2 (green) is the insurance charges model. The dashed arrow shows where they connect.*

## 3.2 Datasets and their roles

Table 2 summarises the two datasets as they appear in the project folder. The figures were counted directly from the files. The disaster file is the project's copy named preprocessed_data.csv. The supplied code does not show how this file was reduced from the original Kaggle download [17], so we describe only what the file contains.

**Table 2: Datasets used in the project**

| Item | US natural disaster declarations [17] | Insurance data [16] |
|---|---|---|
| File used | preprocessed_data.csv | insurance.csv |
| Size | 8,308 rows and 28 columns. Each row is one county in one FEMA declaration. Declaration years 1998 to 2017, 57 states and territories, 380 disaster numbers. | 1,338 rows and 7 columns. Ages 18 to 64. 274 smokers and 1,064 non-smokers. Rows per region: 324 northeast, 325 northwest, 325 southwest and 364 southeast. |
| Relevant columns | state, disasterNumber, incidentType, incidentBeginDate, declarationYear, designatedArea, incidentDuration | age, sex, bmi, children, smoker, region, charges |
| Prediction target | Whether a state has at least one declaration in the next year (yes or no), built by our code. The raw file has no such label. | charges, a number in dollars. The mean is about $13,270. |
| Role in the pipeline | Builds the state-year table and trains the disaster model. Output is a probability for each state. | Trains and tests the charges models. It receives the regional average of the disaster probabilities. |

The two datasets cannot be joined row by row, because the insurance data has no state, county or date. They are therefore used for separate tasks. Their only link is the region name, which is explained in Section 3.5.

## 3.3 Path 1: Disaster risk model

### *Cleaning and event table*

The raw file has one row for each county named in a declaration, so one disaster appears many times. The code collapses these rows to one event for each state and disaster number. It takes the year from the incident start date, or from the declaration year if the start date is missing. Only the 50 states and DC are kept, because territories do not fit the four insurance regions. Years before 1999 are removed, because 1998 holds only a part of the year. This gives 362 state-level events.

Event counts in the file become very small after 2012, with 1 to 10 events a year. The code treats this as a sign that the file is incomplete and not that disasters stopped. It therefore uses 2011 as the last reliable year. The numbers of events by year are 36 in 1999, 28 in 2000 and 27 in 2011, with only 10 in 2012.

### *Features and label*

The code builds a state-year table. For each state and each as-of year t from 2000 to 2010, it computes features using only events up to year t. The label is 1 if the state has at least one event in year t+1. This keeps future information out of the features. The table has 51 units for 11 years, which gives 561 rows. About 35.8% of them are positive.

There are 15 features. They count events in the last 1, 3 and 5 years and in total. They also give the years since the last event (limited to 10), the historical rate of event years, the number of event years in the last five years, and the average number of designated areas and

the average duration of recent events. Finally, they count events of six types in the last five years: severe storm, hurricane, fire, flood, severe ice storm and tornado.

### *Training, testing and models*

The evaluation is a walk-forward test. For each test year from 2006 to 2011, the models train on all earlier target years and predict that year. This gives 306 test rows, of which 100 are positive and 206 are negative. Three models are compared: Logistic Regression with standard scaling, Random Forest with 300 trees, and XGBoost with 300 trees and a depth of 3. All use a seed of 42. The three models were chosen to compare a simple linear model with two tree-based models that can capture non-linear patterns.

A fourth row, called the historical-rate reference, is a baseline. It uses only the state's past rate of event years as the predicted probability. A learned model should beat this reference to justify its extra features. The metrics are accuracy, precision, recall, F1 and ROC-AUC, with a probability threshold of 0.50 for the first four.

### *Selection and risk output*

The learned model with the best ROC-AUC, rounded to three decimals, is chosen. This was Random Forest. It is refitted on all 561 rows. The code then computes the features for every state as of 2011 and predicts the probability of a declaration in 2012. States are grouped into Low, Medium and High using the one-third and two-thirds points of these 51 probabilities (0.236 and 0.369). The categories are therefore relative to other states and not fixed levels of danger. The importance of each feature in the final model is also saved.

## 3.4 Path 2: Insurance charges model

The region names are cleaned to lower case, and the column disaster_risk is added by looking up the regional average from Path 1. The numbers age, BMI, children and disaster_risk are standardised, and the categories sex, smoker and region are one-hot encoded. Table 3 lists the feature sets that are compared.

**Table 3: Feature sets for the insurance models**

| Set | Inputs | Purpose |
|---|---|---|
| A | age, BMI, children, sex, smoker, region | Baseline model without disaster information. |
| B | Set A plus disaster_risk | Disaster-aware model. Tests whether the risk column helps. |
| C | age, BMI, children, sex, smoker, disaster_risk (no region) | Makes risk the only location signal. Used for the what-if curve. |
| D | age, BMI, children, sex, smoker | Used only in a diagnostic check of regional cost levels. |

Three model families are compared: Linear Regression, Random Forest (300 trees, minimum leaf size 3) and XGBoost (400 trees, depth 2, learning rate 0.03, minimum child weight 5, L2 penalty 5, subsample 0.9 and column sample 0.9). The comparison uses 5-fold cross-validation repeated 3 times, which gives 15 fold results, and all models and sets use the same splits. The metrics are MAE, RMSE and R-squared, which are the mean absolute error, the

root mean squared error and the share of variation explained. The family with the lowest RMSE on set A is selected, which was XGBoost, and sets A, B and C are refitted on all 1,338 rows.

To test whether the disaster feature helps, the code compares sets B and A on the same folds. It computes the RMSE gain for each fold and its standard error. A result counts as a real gain only if the average gain is more than two standard errors. The code also checks how many distinct risk values exist in each region, and it compares actual and predicted charges by region using set D.

### 3.5 Risk assessment and estimation logic

The state-to-region rule in config.py places every state in exactly one of four regions: northeast, southeast, southwest or northwest. The comment in the code describes it as roughly east or west of the Plains and north or south of about 37 degrees north. For example, Ohio, Michigan, Illinois and Minnesota are in the northeast group, and California, Texas and Hawaii are in the southwest group. This is a rule written by us and not a machine learning step. A different rule would give different regional averages.

The regional risk is the simple average of the 2012 probabilities of the states in that region. The averages are 0.288 for the northeast, 0.262 for the northwest, 0.461 for the southeast and 0.301 for the southwest.

There is no premium formula in the project. When a user asks for an estimate, the application returns two predicted charges. The baseline estimate comes from Model A, which uses the region. The disaster-aware estimate comes from Model B, which uses the region and its risk value. Negative predictions are set to zero. The difference between them is shown to the user. The explanation text in the interface changes with the cross-validation evidence. Because the evidence showed no measurable gain, the text tells the user to treat the gap as model noise and not as a disaster effect.

### 3.6 How the interface connects to the models

The web application uses Flask for the server and a single HTML page with plain JavaScript for the interface. At start-up, the server loads the saved insurance models, the state and region risk tables and results.json from the artifacts folder. If a file is missing, it asks the user to run train.py first. The server has four routes: one returns the state and region data, one returns the saved results, one produces an estimate and one produces the what-if curve.

The workflow for one estimate is as follows. The user chooses a state and enters age, BMI, number of children, sex and smoking status. The page sends them to the estimate route. The server checks that age is from 18 to 100, BMI from 10 to 60 and children from 0 to 10, and it adds a warning if the age is above 64. It maps the state to its region and looks up the state risk and the region risk. It then runs Model A and Model B, builds the explanation and returns everything to the page, which shows the result. For the what-if slider, the server runs Model

C for 101 risk values from 0 to 1 and returns the curve together with the four risk levels seen in training.

### 3.7 Tools and libraries

The project is written in Python 3. It uses pandas and NumPy for data handling, SciPy for the rank correlation check, scikit-learn [19] for Logistic Regression, Random Forest, preprocessing and cross-validation, XGBoost [20] for gradient boosting, joblib to save models, and Flask for the web server. If XGBoost is not installed, the code falls back to scikit-learn's histogram gradient boosting. The saved results show that XGBoost was installed for the recorded run.

### 3.8 Limitations of the method

Several limits follow from the design and the evidence. The disaster table is small, with 561 rows and only six test years, and the later years in the file are thinly filled. The disaster model sees only past counts, event types and durations. It has no weather, climate, population or exposure data. The risk shown in the interface is a forecast for 2012 and is not a current forecast.

For the insurance data, the disaster_risk column is a fixed function of region, so Model B holds no information that Model A lacks. The insurance file also has no date, so the 2012 risk value is attached to people whose year is unknown. The XGBoost settings for the insurance model were chosen to lower the cross-validation error, and the same cross-validation is used to report the scores. The scores are therefore estimates from this dataset and not results on an untouched final test set. The region rule is an assumption. Finally, the target is health insurance charges, so the results say nothing about property insurance, claims or disaster losses.

## 4. Results and Discussion

This section first walks through the application in the order a user would use it, and then reports the model results. The values come from the saved file results.json and the project's results note (results_explained.readme), and they were checked against the metric images. Where the sources differ, the difference is stated.

### 4.A User interface and application outputs

A user starts on the Estimator tab, shown in Figure 2. The left panel holds the inputs: state, age, BMI, number of children, sex and smoking status. The example uses a 35-year-old non-smoking woman in Florida with a BMI of 27 and two children. The right panel shows six result cards. Florida belongs to the Southeast region. Its state disaster risk for 2012 is 37%, labelled Medium, and the Southeast region averages 46%, labelled High. The baseline estimate from Model A is $7,487 and the disaster-aware estimate from Model B is $7,372. The difference is -$115. The yellow note explains that adding disaster risk did not

measurably improve cost prediction, so the gap should be read as model noise and not as a disaster effect.

The lower part of the same tab is the what-if panel. The slider changes the risk value and Model C, which has no region input, gives an estimate. At a risk of 46% the Model C estimate is $7,222. The curve is flat on the left and right and drops sharply near 30%. The four dots are the only risk values seen in training, so everything between them is the model's extrapolation. The blue note under the graph says so.

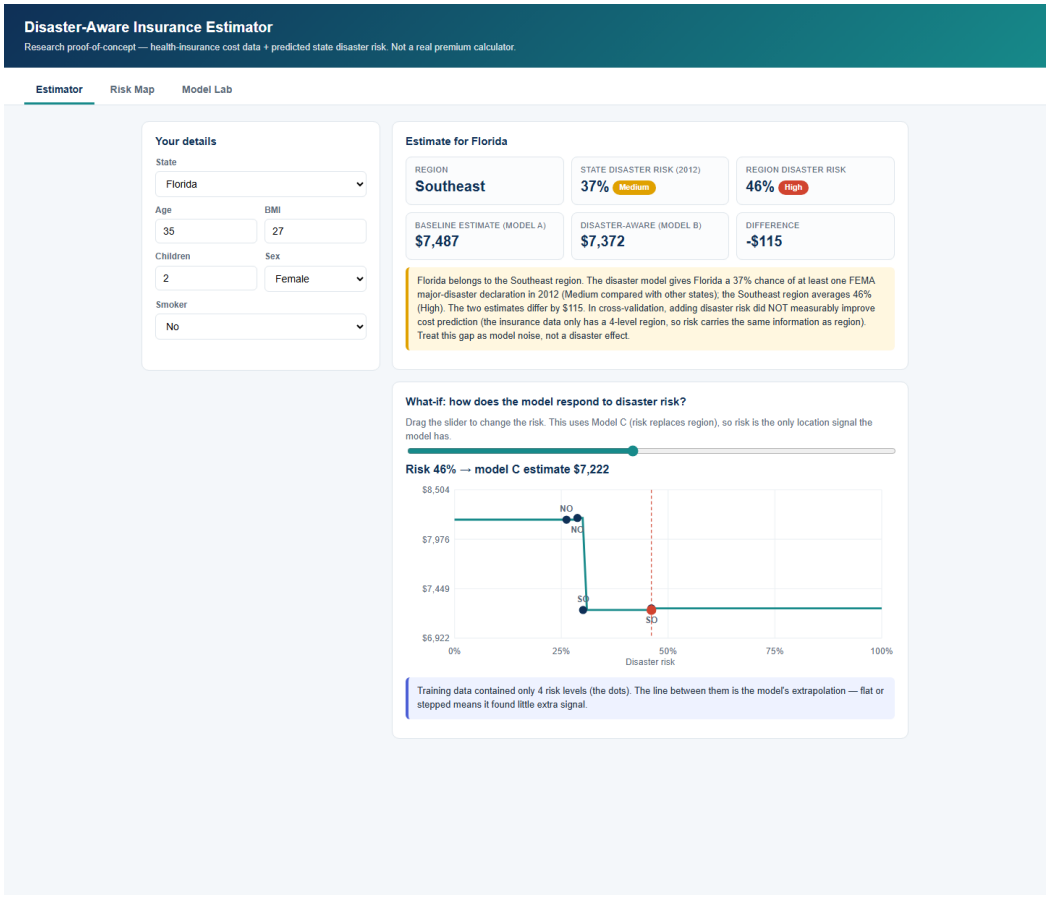



*Figure 2: Estimator tab with the inputs, the six result cards, the explanation note and the what-if panel.*

Figure 3 shows the Risk Map tab. It draws each state as a tile, coloured from light to dark orange as the predicted chance of at least one FEMA major disaster declaration in 2012 rises. Kentucky (66%), Texas (61%), Mississippi (60%), Tennessee (58%) and North Carolina (55%) have the highest values, and Wyoming (15%), New Hampshire (16%) and Minnesota (16%) are among the lowest. Clicking a tile selects that state for the estimator, and Florida is outlined in the figure because it was the selected state. The cards below the map show the regional averages, which are the values passed to the insurance model.

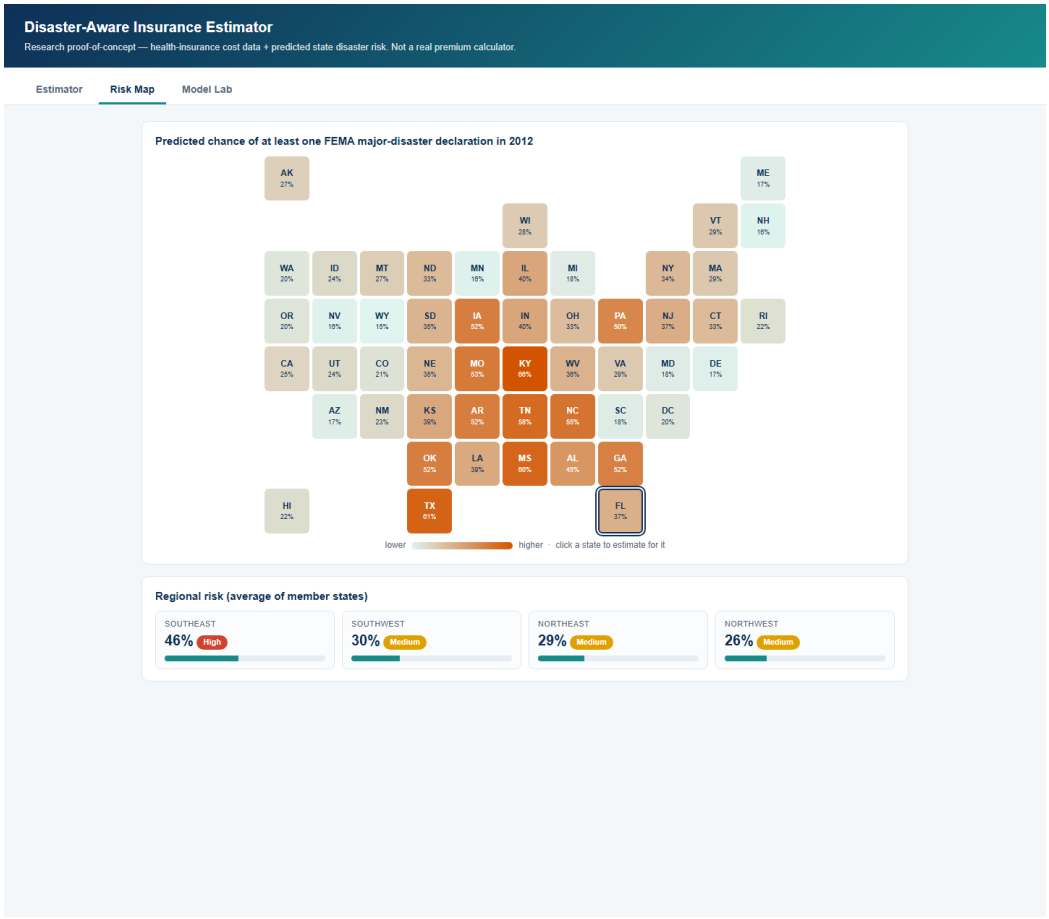



*Figure 3: Risk Map tab with the 2012 declaration probability for each state and the four regional averages.*

Figure 4 shows the Model Lab tab, which holds the evaluation tables. The top card reports the disaster model on 362 events and 561 state-year rows with the four rows of metrics, and a bar list called What the selected model relies on. The middle card reports the insurance models for the nine combinations of family and feature set, followed by the paired comparison of Model B against Model A. The bottom card is titled Why B cannot beat A on this dataset. It states that disaster risk takes only four values, one per region, and lists the regional risk, mean charges and cost relativity. The metrics in this tab are discussed in Section 4.B.

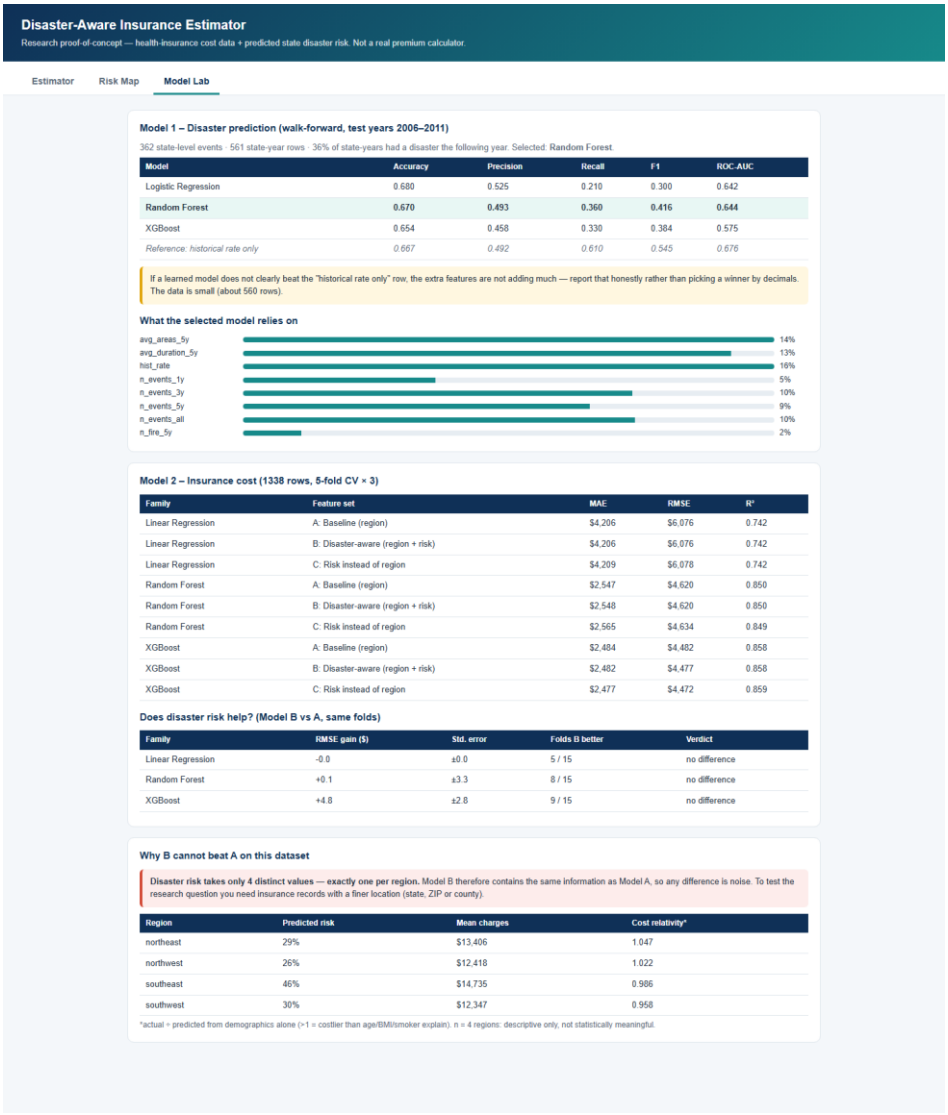



*Figure 4: Model Lab tab with the disaster model table, the insurance model tables, the paired comparison and the regional diagnostic.*

Two small points in the interface deserve a note. The bar list in Figure 4 shows the first eight feature names in alphabetical order, not the eight most important features, so it leaves out two features that matter more than some it shows (Section 4.B). In the what-if graph, the dots are labelled with the first two letters of the region name, so the northeast and northwest are both shown as NO and the southeast and southwest are both shown as SO.

## 4.B Model performance and metrics

### *Disaster model*

The disaster model predicts a yes or no outcome, so we use classification metrics. Accuracy is the share of all state-years that were classified correctly. Precision is the share of predicted disaster years that were real disaster years. Recall is the share of real disaster years that the model found. F1 combines precision and recall. ROC-AUC measures how well the model ranks disaster years above other years, and 0.5 means no better than chance. Figure 5 is the metrics image from the application, and Table 4 repeats its values with the threshold at 0.50.

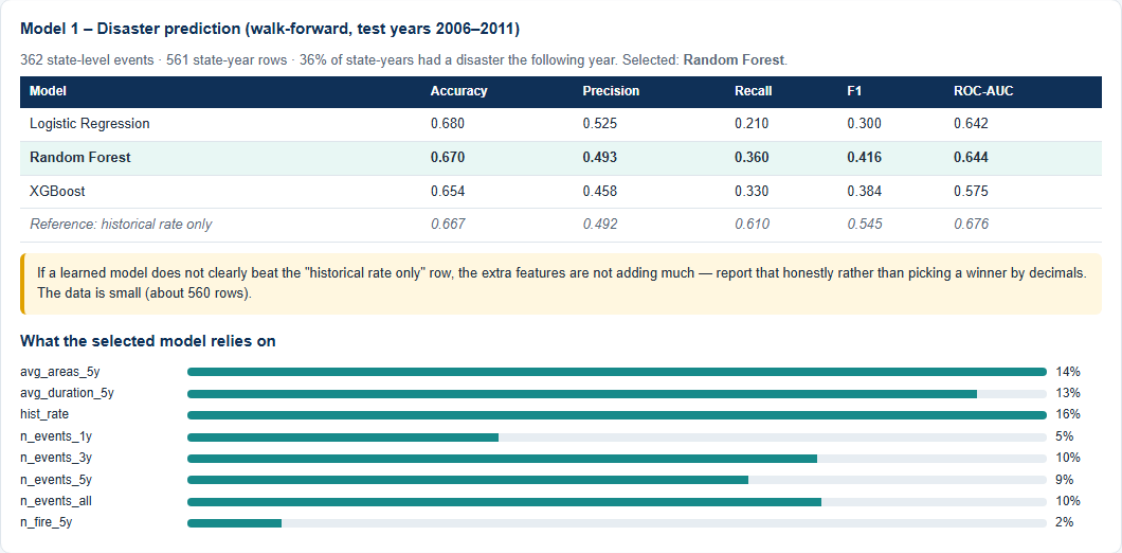



*Figure 5: Disaster model results from walk-forward validation on test years 2006 to 2011, with the feature bars shown by the interface.*

**Table 4: Disaster model results (walk-forward, 306 test rows, threshold 0.50)**

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.680 | 0.525 | 0.210 | 0.300 | 0.642 |
| **Random Forest (selected)** | 0.670 | 0.493 | 0.360 | 0.416 | 0.644 |
| XGBoost | 0.654 | 0.458 | 0.330 | 0.384 | 0.575 |
| Historical-rate reference | 0.667 | 0.492 | 0.610 | 0.545 | 0.676 |

The scores are modest, and the evidence shows why. Of the 306 test rows, 206 are negative, so a model that always says no disaster would score 206/306 = 67.3% accuracy. The selected Random Forest scores 67.0%, which is almost the same. It does find some real disaster years, with a recall of 0.36, while the always-no model has a recall of zero. But the accuracy figure alone does not show skill. The historical-rate reference has higher recall (0.61), F1 (0.545) and ROC-AUC (0.676) than the Random Forest. The learned models therefore do not clearly beat the simple reference, and the extra features add little on this data.

The threshold trades one metric for another. Table 5 shows the Random Forest at the default 0.50 and at 0.30, which is reported in the results note. The lower threshold finds more disaster years but gives more false alarms, so accuracy and precision fall. ROC-AUC does not depend on the threshold. The values saved and shown in the interface use 0.50.

**Table 5: Selected Random Forest at two thresholds**

| Threshold | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| 0.50 (reported) | 0.670 | 0.493 | 0.360 | 0.416 |
| 0.30 | 0.578 | 0.414 | 0.700 | 0.520 |

Several reasons for the modest scores are supported by the data. The panel is small, and each test year has only 51 rows. The share of disaster years changes from year to year, from 21.6% in 2009 to 43.1% in 2007, so one threshold cannot suit every year. The model sees only past

counts and event summaries, and whether a declaration happens in a given year depends on many other things. These are explanations that fit the evidence, and we have not tested whether extra inputs such as weather or population would improve the scores.

The feature bars in Figure 5 should be read with care. The interface shows the first eight names in alphabetical order. The saved importance values, from the final Random Forest, rank the top features as follows: hist_rate 16.0%, avg_areas_5y 13.8%, avg_duration_5y 12.7%, n_events_all 10.2%, n_events_3y 10.1%, n_events_5y 9.0%, n_severe_storm_5y 7.5% and years_since_last 5.9%. The figure leaves out the last two and shows n_events_1y (5.0%) and n_fire_5y (1.5%) instead. The values that are shown match the saved file. Importance values of this kind describe what the model used and do not show what causes disasters.

### *Insurance charges model*

The insurance model predicts a dollar amount, so we use regression metrics. MAE is the average size of the error in dollars. RMSE is also in dollars but punishes large errors more. R-squared is the share of variation in charges explained by the model, where 1 is perfect and 0 equals predicting the average. R-squared is not accuracy. Figure 6 is the metrics image from the application, and Table 6 repeats the values.



**Model 2 – Insurance cost (1338 rows, 5-fold CV × 3)**

| Family | Feature set | MAE | RMSE | R² |
|---|---|---|---|---|
| Linear Regression | A: Baseline (region) | $4,206 | $6,076 | 0.742 |
| Linear Regression | B: Disaster-aware (region + risk) | $4,206 | $6,076 | 0.742 |
| Linear Regression | C: Risk instead of region | $4,209 | $6,078 | 0.742 |
| Random Forest | A: Baseline (region) | $2,547 | $4,620 | 0.850 |
| Random Forest | B: Disaster-aware (region + risk) | $2,548 | $4,620 | 0.850 |
| Random Forest | C: Risk instead of region | $2,565 | $4,634 | 0.849 |
| XGBoost | A: Baseline (region) | $2,484 | $4,482 | 0.858 |
| XGBoost | B: Disaster-aware (region + risk) | $2,482 | $4,477 | 0.858 |
| XGBoost | C: Risk instead of region | $2,477 | $4,472 | 0.859 |

**Does disaster risk help? (Model B vs A, same folds)**

| Family | RMSE gain ($) | Std. error | Folds B better | Verdict |
|---|---|---|---|---|
| Linear Regression | -0.0 | ±0.0 | 5 / 15 | no difference |
| Random Forest | +0.1 | ±3.3 | 8 / 15 | no difference |
| XGBoost | +4.8 | ±2.8 | 9 / 15 | no difference |

*Figure 6: Insurance model results from 5-fold cross-validation repeated three times, and the paired comparison of Model B against Model A.*

**Table 6: Insurance model cross-validation averages (15 folds)**

| Family | Feature set | MAE | RMSE | R-squared |
|---|---|---|---|---|
| Linear Regression | A: baseline (region) | $4,206 | $6,076 | 0.742 |
| Linear Regression | B: region + risk | $4,206 | $6,076 | 0.742 |
| Linear Regression | C: risk instead of region | $4,209 | $6,078 | 0.742 |
| Random Forest | A: baseline (region) | $2,547 | $4,620 | 0.850 |

| Family | Feature set | MAE | RMSE | R-squared |
|---|---|---|---|---|
| Random Forest | B: region + risk | $2,548 | $4,620 | 0.850 |
| Random Forest | C: risk instead of region | $2,565 | $4,634 | 0.849 |
| **XGBoost** | A: baseline (region) | $2,484 | $4,482 | 0.858 |
| **XGBoost** | B: region + risk | $2,482 | $4,477 | 0.858 |
| **XGBoost** | C: risk instead of region | $2,477 | $4,472 | 0.859 |

XGBoost has the lowest error, so it was selected. With set A, its average MAE is $2,483.74, its RMSE is $4,481.64 and its R-squared is 0.8579. The spread across folds is about $221.84 for MAE, $528.41 for RMSE and 0.04 for R-squared, so scores differ from fold to fold. The results note also says that the XGBoost settings were changed to a shallower, more regularised version. It reports that set A improved from an MAE of $2,542.63, an RMSE of $4,591.65 and an R-squared of 0.8527 before the change. The before-change numbers are not stored in the saved outputs, so we could not check them. Because the same folds were used to choose and to report, these gains are estimates and not proof of performance on new data.

The three feature sets give almost the same scores. Table 7 shows the paired comparison of set B against set A. The gain is the drop in RMSE when the risk column is added. For XGBoost it is $4.80, and its standard error is $2.80, which is below the two standard errors needed to call it a gain. B was better on only 9 of the 15 folds. For the other two families the gain is close to zero. The verdict for all three is no difference. We therefore cannot say that disaster risk improves the prediction of insurance charges.

**Table 7: Paired comparison of Model B against Model A on the same folds**

| Family | RMSE gain of B | Std. error | Folds B better | Verdict |
|---|---|---|---|---|
| Linear Regression | -$0.0 | $0.0 | 5 of 15 | No difference |
| Random Forest | +$0.1 | $3.3 | 8 of 15 | No difference |
| XGBoost | +$4.8 | $2.8 | 9 of 15 | No difference |

This result is expected from the design. Each region has exactly one risk value, so set B contains the same information as set A, and set C only recodes the region. Table 8 shows the regional diagnostic. Cost relativity is the actual total charges divided by the charges predicted from age, sex, BMI, children and smoking alone, so a value above 1 means the region is costlier than those inputs explain. The southeast has the highest risk (46%) and a relativity of 0.986, and the northeast has a lower risk (29%) and a relativity of 1.047. The rank correlation between risk and relativity is -0.6. With only four regions this is a description and not a statistical result, and we do not read it as an effect.

**Table 8: Regional diagnostic (set D predictions)**

| Region | Predicted risk | Mean charges | Cost relativity |
|---|---|---|---|
| Northeast | 28.8% | $13,406 | 1.047 |
| Northwest | 26.2% | $12,418 | 1.022 |

| Region | Predicted risk | Mean charges | Cost relativity |
|---|---|---|---|
| Southeast | 46.1% | $14,735 | 0.986 |
| Southwest | 30.1% | $12,347 | 0.958 |

Figure 7 is the what-if graph and Figure 8 is the risk map from the metric photos. The graph shows Model C as a step. The estimate is higher for the two regions with lower risk (northeast and northwest) and lower for the two regions with higher risk (southwest and southeast). At a risk of 46% the estimate is $7,222. The sharp step falls between the 29% and 30% points. This is the model separating the four regions. It is not a smooth effect of risk. Figure 8 shows the same risk map as Figure 3, so we do not describe it again.

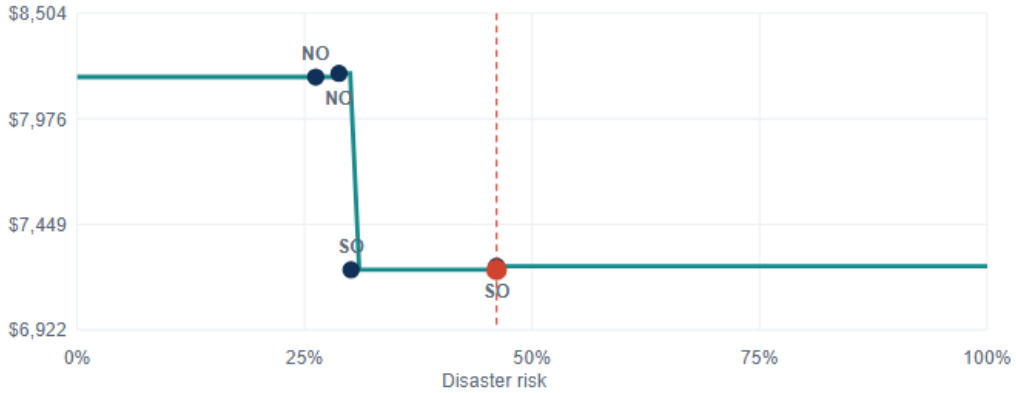



*Figure 7: What-if curve from Model C for the sample profile. The dots are the four risk levels seen in training.*

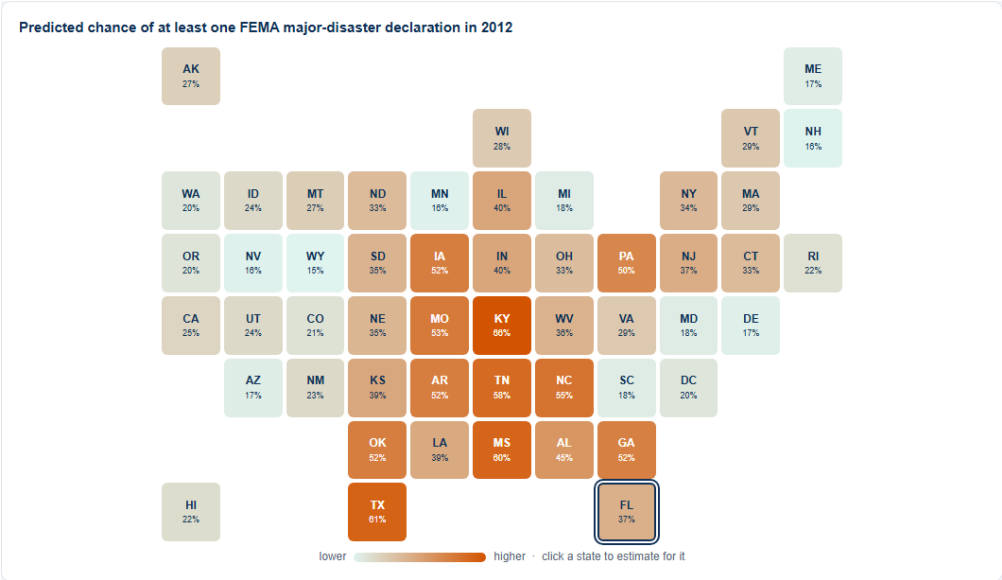



*Figure 8: Predicted chance of at least one FEMA major disaster declaration in 2012 for each state (the same map as in the Risk Map tab).*

**Points to confirm.** The results note gives R-squared of 0.8578 for set B and 0.8586 for set C with XGBoost, while results.json gives 0.8582 and 0.8585. The differences are at the fourth decimal place and do not change any conclusion. We use three decimals in Table 6, where all sources agree. The exact four-decimal values should be confirmed by the team before final submission.

### 4.C Comparison with published work

Direct number-to-number comparison needs the same task, data and evaluation setup. Our project does not share these with any of the reviewed papers, so Table 9 compares methods, scope and limits. We do not claim that our project performs better or worse than any published work.

**Table 9: Comparison with related papers**

| Paper | Task and data | Reported evaluation | Relation to our project |
|---|---|---|---|
| [1] | Multi-class disaster type, synthetic severity and claims, FEMA records from 1953 to 2025, premium multiplier. | Temporal hold-out accuracy 53.86%, random split 92.24%. | We predict a yes or no declaration, not the type, and we use no synthetic loss. Same data family. Accuracy values are not comparable. We share the lesson that time-based tests are harder than random splits. |
| [2], [6], [7] | Flood claims from NFIP data, with terrain, rain and social inputs. | R-squared from 0.5 to 0.95 depending on the scale [7]. R-squared 0.807 in a regional study [6]. | We have no claims data and no flood-specific inputs. Their results show that finer location and weather inputs matter. |
| [8], [11], [12] | Hurricane losses or claims using building-level data, simulated events or policy data. | MAE of 11.3% at ZIP level [8]. Claim counts within 2% to 12% [12]. | These use detailed property or storm data. Ours uses state-level counts only, so we cannot match their scope. |
| [10] | Review of 94 hurricane risk studies. | Not applicable. | Our disaster model is based on counts of declarations and not on damage, which is the weakness the review points out. |
| [3], [9], [14], [15] | How real home insurance prices relate to risk, regulation and credit scores. | Not applicable. | We do not model real prices. These studies show that useful risk pricing needs location detail finer than our four regions. |

Overall, our disaster model reached an ROC-AUC of 0.644 and did not beat a simple historical-rate reference on most measures. Our insurance model fitted the charges data well, with an R-squared of about 0.858, but the disaster feature did not add information. A high score on one dataset does not show that a model is reliable in the real world. Paper [1] reports the same pattern, where strong random-split scores fell sharply under a time-based test. Our results support a careful reading: the pipeline works as a demonstration, and its main finding is that region-level disaster risk cannot be tested properly with this insurance data.

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# 6. Code and Dataset

The three GitHub links below were opened without logging in on 5 November 2026, and each one returned its page or file. GitHub did not ask for an account.

**Table 10: Code and dataset links**

| Item | Link | What it contains and how to access it |
|---|---|---|
| Project source code on GitHub | https://github.com/03AJ03/AF_MiniProject | The public repository. Its top level lists a DATASETS folder, a disaster_insurance folder, a LICENSE file, app.py, disaster_model.py, insurance_model.py, index.html and disaster_insurance.zip. Open the page and download the code or clone it. |
| Dataset location on GitHub (insurance data) | https://github.com/03AJ03/AF_MiniProject/blob/main/DATASETS/insurance.csv | The insurance file with 1,338 rows. Its content is the same as the copy used in our code. Use the Raw button on the page to download it. |
| Dataset location on GitHub (disaster data) | https://raw.githubusercontent.com/03AJ03/AF_MiniProject/refs/heads/main/DATASETS/preprocessed_data.csv | The disaster declarations file with 8,308 rows. It is a direct download and is identical to the copy used in our code. |
| Original insurance dataset | https://www.kaggle.com/datasets/mirichoi0218/insurance | The Kaggle source of the insurance data [16]. |
| Original natural disaster dataset | https://www.kaggle.com/datasets/headsortails/us-natural-disaster-declarations | The Kaggle source of the disaster declarations [17]. |

To run the project, install the packages in requirements.txt (pandas, numpy, scipy, scikit-learn, xgboost, flask and joblib). Then run python train.py to train both models and write the artifacts folder, and run python app.py to open the interface at http://127.0.0.1:5000. These steps are written in the README that is inside the disaster_insurance project folder. The Kaggle pages were not opened during this check, so any sign-in requirement on Kaggle is not confirmed here.